

## Part VI: The Algebra of Relational Coupling – Foundations of Dynamics and Gauge Emergence

**Abstract:** This part establishes the algebraic and dynamic foundation for the coupling operator  $\hat{J}_{ij}$  in Vibrational Geometrodynamics. We define the algebra of this operator as acting in the spin space of nodes, derive an effective action for the coupling field, and explore how non-local terms in this action can lead to the emergence of gauge-like structures. The model provides a bridge between the quantum regulation of coupling via spin (Part V) and the topological description of extended objects (Part VII).

### VI.1 The Operator Algebra of the Coupling $\hat{J}_{ij}$

The coupling operator  $\hat{J}_{ij}$  treats the link between nodes  $i$  and  $j$  as an independent quantum degree of freedom. It acts in the tensor product space of the spins of the two nodes,  $\mathcal{H}_i \otimes \mathcal{H}_j$ . For spin -1/2 nodes, the most general Hermitian and SU(2)-rotation invariant form is:

$$\hat{J}_{ij} = J_0(v, r_{ij}) \mathbb{I}_i \otimes \mathbb{I}_j + J_1(v, r_{ij}) \vec{\sigma}_i \cdot \vec{\sigma}_j$$

where:

- $J_0$  and  $J_1$  are real functions of the background geometric potential  $v$  and an effective distance  $r_{ij}$ .
- $\vec{\sigma}_i$  are the Pauli matrices for node  $i$ .
- The term  $\vec{\sigma}_i \cdot \vec{\sigma}_j$  implements spin-dependent filtering: it takes the value +3 for parallel spins (triplet) and -3 for anti-parallel spins (singlet), thus modulating the effective coupling by a factor of  $\pm 3J_1$ .

**Fundamental commutation relations:** We postulate that  $\hat{J}_{ij}$  does not fully commute with the spin operators of its nodes, reflecting the deep coupling between geometry (coupling) and quantum content (spin). A minimal relation preserving the relational character could be:

$$[\hat{J}_{ij}, \hat{S}_i^a] = i\epsilon^{abc} \lambda J_1 \sigma_i^b \otimes \sigma_j^c$$

where  $\lambda$  is a small dimensionless coupling constant, and  $\epsilon^{abc}$  is the Levi-Civita tensor. This relation suggests that the act of modifying the coupling perturbs the relational spin state and vice versa, implementing feedback at the operator level.

### VI.1.1: The Geometric Background Potential ( $\nu$ )

To close the operator algebra, it is necessary to specify the origin of the parameters  $J_0$  and  $J_1$ . These are functions of a geometric background potential, denoted by  $\nu$ .

**Definition and Significance:** In the VG formalism,  $\nu$  is a functional of the primary geometry field  $\Phi$  and the node distribution. It encodes the primordial geometric information of the network in the absence of spin-regulated quantum coupling. More precisely:

$$\nu \equiv \nu[\Phi, \nabla\Phi, \{r_{ij}\}]$$

where  $\{r_{ij}\}$  denotes a measure of nodal separation in the reference configuration.

Its role in the dynamics is twofold:

1. Source of the "bare" coupling: It generates the base values  $J_0(\nu, r_{ij})$  and  $J_1(\nu, r_{ij})$  that would characterize the coupling in a world without relevant spin or quantum fluctuations. It is analogous to the spring constant of an arc before it is loaded or interacts with anything else.
2. Driver for the equations of motion: As it appears in the dynamical equation from Part V, its derivative provides the geometric restoring force that tends to bring the coupling back to its background-determined equilibrium value.

In essence,  $\nu$  is the interface between the static (or quasi-static) geometry of the network and the emergent quantum dynamics of the coupling. A complete development of its functional form is left for future work, as this would require a full specification of the dynamics of the  $\Phi$  field. For present purposes, it is sufficient to consider it as the parametric source for the algebraic structure of  $\hat{J}_{ij}$

## VI.2 The Effective Action and Non-Local Dynamics

To obtain the effective classical dynamics of the coupling, we move from the operator description to a continuous field one. We define  $J_{ij}(t)$  as the expectation value  $\langle \hat{J}_{ij} \rangle$  and consider an effective action that includes both the potential energy from Part V and kinetic and non-local terms:

$$S[J] = \int dt \sum_{\langle ij \rangle} \left[ \frac{1}{2g^2} \left( \frac{dJ_{ij}}{dt} \right)^2 - V(J_{ij}) \right] - \sum_{\langle ijk \rangle} \frac{K}{2} (J_{ij} - J_{jk})^2$$

where:

- The first term is the kinetic energy of the temporal fluctuations of the coupling, with  $g$  a characteristic propagation speed.
- $V(J) = \frac{\partial v}{\partial J} - \mu \langle \vec{S}_i \cdot \vec{S}_j \rangle J + \frac{\xi}{2} J^4$  is the local potential derived from the equation in Part V.
- The crucial term is the last one: the "hopping" or non-local rigidity term  $K$ . This penalizes large differences in coupling between adjacent links ( $ij$  and  $jk$ ) that share a common node, favoring homogeneity and introducing spatial correlations in the network.

The Euler-Lagrange equation for this action gives the generalized dynamic equation:

$$\frac{1}{g^2} \frac{d^2 J_{ij}}{dt^2} = - \frac{\partial V}{\partial J_{ij}} + K \sum_{k \in \mathcal{N}(i,j)} (J_{jk} + J_{ki} - 2J_{ij})$$

where the sum is over nodes  $k$  neighboring the pair  $i, j$ . The term proportional to  $K$  is a discrete Laplacian-type operator that leads to the diffusion and dispersion of perturbations in the coupling across the entire network. This is the fundamental mechanism for the propagation of coherence and effective interactions.

### VI.3 Emergence of Gauge Connections from Phase Fluctuations

The rigidity term  $K$  is analogous to the kinetic term for a gauge connection in a discrete theory. To see this, we separate the coupling into amplitude and phase:  $J_{ij} = \rho_{ij} e^{i\theta_{ij}}$ . Considering the case where the amplitudes  $\rho_{ij}$  are nearly constant (in a certain network phase), the dynamics are dominated by phase fluctuations  $\theta_{ij}$ .

We define a U(1) connection on edges:

$$A_{ij} \equiv \frac{1}{\theta_0} \theta_{ij}$$

where  $\theta_0$  is a reference angle. The hopping term  $K$  can be rewritten, for small phase fluctuations, as:

$$K(J_{ij} - J_{jk})^2 \approx \text{const.} \times K\rho^2(\theta_{ij} - \theta_{jk})^2 = \tilde{K}(A_{ij} + A_{jk} + A_{ki})^2$$

the last equality highlighting that the sum of phases around a triangle  $i \rightarrow j \rightarrow k \rightarrow i$  is the curvature associated with the connection.

Wilson Loop and force fields: For a closed loop  $C$  in the network, the Wilson Loop  $W_C$  is:

$$W_C = \prod_{\langle ij \rangle \in C} \frac{J_{ij}}{|J_{ij}|} = \exp \left( i \sum_{\langle ij \rangle \in C} \theta_{ij} \right) \approx \exp \left( i\theta_0 \sum_{\langle ij \rangle \in C} A_{ij} \right)$$

In the continuum limit of a dense network, the sum  $\sum_C A_{ij}$  becomes the line integral  $\oint_C A_\mu dx^\mu$ , and the effective action for  $A_{ij}$  can tend towards a Maxwell-Yang-Mills type action  $S \sim \int d^4x F_{\mu\nu} F^{\mu\nu}$ , where  $F_{\mu\nu}$  is the curvature.

Implication: Photons (or potentially other gauge bosons) appear in VG as collective vibration modes of the phase of relational couplings  $J_{ij}$ . The U(1) gauge symmetry is an emergent symmetry of the phase dynamics in the low-energy limit.

## VI.4 Simulability and Open Questions for Research

**1. Numerical simulation:** The model is suitable for simulations on finite lattices. Essential parameters are:  $J_0, J_1, \mu, \xi, g, K$ . The following can be explored:

- **Phase diagram:** Searching for a transition between an "insulating" phase (weak, uncorrelated couplings, small  $K$ ) and a "coherent" or "conducting" phase (strong couplings, aligned phases, large  $K$ ).

- **Emergence of topological excitations:** Stable solitonic configurations of the  $J_{ij}$  and  $\langle \vec{S}_i \cdot \vec{S}_j \rangle$  fields that propagate – the candidate for the VG electron.

- **Gauge-type correlations:** Measuring the average value of Wilson Loops  $\langle W_C \rangle$  to detect a transition between a confining and a deconfining phase for the emergent gauge excitations.

### 2. Theoretical open questions:

- **Extension of symmetry groups:** If the nodal spin space allows a larger symmetry (e.g.,  $SU(N)$ ), the algebra of  $\hat{J}_{ij}$  can be extended to  $\hat{J}_{ij} = \sum_a T_i^a \otimes T_j^a$ , leading to the emergence of non-Abelian gauge fields.

- **Coupling to gravity:** How do phase ( $A_{ij}$ ) and amplitude ( $\rho_{ij}$ ) fluctuations couple to the underlying vibrational dynamics ( $q_i, p_i$ )? The amplitude could be directly related to the metric or the gravitational connection.

- **Origin of time:** The time derivative in the action  $S[J]$  presupposes a fundamental time. Could this be derived from an additional dimension in the network or from an ordering parameter in the coupling configuration?

## VI.5 Conclusion and Preparation for Part VII

This part has consolidated VG as a dynamic theory of relations. We have defined:

- The coupling as a quantum operator with its own algebra.
- Its dynamics through an effective action with non-local terms ( $K$ ).
- The emergence of gauge structures from the phase fluctuations of the coupling.

This mathematical apparatus is the perfect tool to address, in Part VII, the complex problem of the black hole. We will be able to model:

- The horizon as a topological constriction where the rigidity parameter  $K$  becomes negative or nullified.
- The interior as a sub-network with its own dynamics (*strong*  $J_{\text{int}}$ ), weakly connected to the rest of the universe by an extremely small or zero value of  $K_{\text{horizon}}$ .
- The quantum black-to-white hole transition as a dynamic reconfiguration of this constriction.

Final bridge: The algebra and dynamics described here provide the precise language in which the physical poetry of the "hidden palace" can be formulated.

Authors: Physicist Rodica Pop & DeepSeek AI Assistant